

1-Bit Half Adder Using Dataflow Modeling

Aim

To design and verify a 1-bit half adder in Verilog using dataflow modeling.

Code

```
module half_adder(  
    input A,  
    input B,  
    output Sum,  
    output Carry  
);  
  
    assign Sum = A ^ B;  
    assign Carry = A & B;  
  
endmodule
```

Truth Table / Observation

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Observation: All 4 input combinations were simulated and matched the expected half-adder behavior. The simulation generated `wave.vcd` and completed at 40 ns.

Conclusion

The 1-bit half adder was successfully implemented using Verilog dataflow modeling. XOR generates the sum and AND generates the carry.